

Cisco:

Cisco's name is derived from the city San Francisco. Cisco Systems designs and sells communications and networking technology as well as services. By early 2000, Cisco was the third most valuable company in the entire world, ranking just behind Microsoft and General Electric Company.



Cisco Products



Cisco ASA:

- o The ASA in Cisco ASA stands for Adaptive Security Appliance, security device.
- o Cisco started in the firewall business with a legacy firewall called **Centri** Firewall.
- o Then released very popular firewall called **PIX** (Private Internet Exchange) Firewall.
- o Then introduced the Cisco legacy **IDS** and **Intrusion prevention system** 4200 Series.
- o Cisco introduced the Cisco ASA line of security appliances in May of 2000.
- o ASA combines functionality from PIX, VPN concentrator, and IDS product lines.
- o Through version 7 of the software, the PIX and ASA used the same software.
- o Beginning with version 8, the software was split with different software images.
- o ASA family ranges from ASA 5505 small office/home office appliance to ASA 5585-X.
- o Cisco PIX, which provided firewall and NAT functions ended sale on 28 July 2008.
- o Cisco IPS 4200 Series, which worked as intrusion prevention systems (IPS).
- o Cisco VPN 3000 Series Concentrators, which provided virtual private networking (VPN).
- o ASA is unified threat management device, combining several network security functions.
- o The Cisco ASA Firewall is a security device that combines many things in one box.
- o The ASA combines Firewall, Antivirus, Intrusion Prevention, and VPN capabilities.
- o Beyond being a Firewall, the Cisco ASA can do Antivirus, Antispam & IDS/IPS engine.
- o ASA also do VPN Device, SSL Device and Content Inspection and many more things.
- o It is a network security device that monitors incoming and outgoing network traffic.
- o Decides whether to allow or block specific traffic based on defined set of security rules.
- o Cisco ASA firewall have been first line of defense in network security for over 15 years.
- o ASA has become one of most widely used firewall/VPN solutions for small to medium.
- o ASA software is based on Linux, it runs single Executable and Linkable Format program.
- o Cisco ASA Firewall software has a similar interface to the Cisco IOS software on routers.
- o There is a command line interface that can be used to query operate or configure device.
- o Configuration is initially in memory as running-config but would be saved to flash memory.



Cisco ASA Features:

- o The Cisco ASA Firewall is State-Full Firewall, maintain the state of connection.
- o User Base Authentication using Authentication Authorization & Accounting.
- o Cisco ASA Protocol and application inspection deeper inspection of application.
- o Cisco ASA Firewall Modular Policy Framework (MPF) Inspection of connection.
- o VPN Cisco ASA support IPSec, SSL PPTP protocols for Virtual Private Network.
- o One of the main Cisco ASA firewall feature is virtual firewall or security context.
- o Web Base management using Graphical User Interface GUI via ASDM interface.
- o Cisco ASA firewall can be configured as Transparent Firewall Layer 2 firewall.
- o Cisco ASA firewall support Full Failover active-standby and active-active failover.
- o Cisco ASA Firewall also support Internet Protocol Version 6 (IPv6) routing as well.
- o Another feature clustering enables to group multiple appliances as single appliance.
- o VPN Load Balancing enable multiple remote VPN servers to appear as single server.

Firewall Services:

- o User Base Authentication.
- o State-Full Firewall.
- o Protocol and application inspection.
- o Modular Policy Framework.
- o VPN Support.
- o Virtual Firewalls.
- o Web Base management.
- o Transparent Firewall.
- o State Full Failover.
- o Internet Protocol V6.
- o Clustering feature.
- o VPN Load Balancing.
- o Network Address Translation (NAT).
- o Layer 3 or Layer 2 implementation.
- o Botnet Traffic Filtering.
- o Advanced Malware Protection.
- o Reputation-based URL filtering.
- o Dynamic Host Configuration Protocol.

Cisco ASA Model:

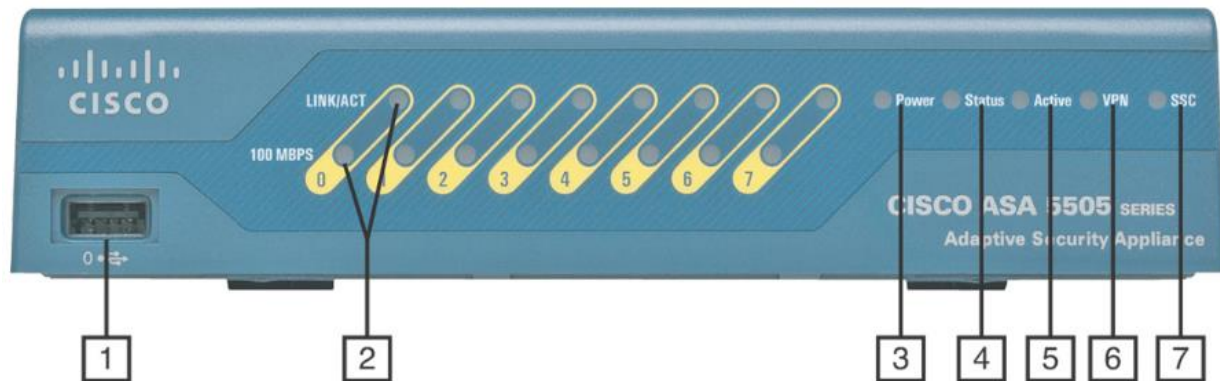
There are several Cisco ASA 5500 and ASA 5500-X Series Next-Generation Firewall models.

Models	Description
Cisco ASA 5505	Small offices and branch offices
Cisco ASA 5510	Small offices and branch offices
Cisco ASA 5512-X	Small offices and branch offices
Cisco ASA 5515-X	Small offices and branch offices
Cisco ASA 5520	Medium-sized offices Internet-edge security appliances
Cisco ASA 5525-X	Medium-sized offices Internet-edge security appliances
Cisco ASA 5540	Medium-sized offices Internet-edge security appliances
Cisco ASA 5545-X	Medium-sized offices Internet-edge security appliances
Cisco ASA 5550	Large enterprise Internet-edge security appliances
Cisco ASA 5555-X	Medium-sized offices Internet-edge security appliances
Cisco ASA 5585-X SSP-10	Data center and large enterprise networks
Cisco ASA 5585-X SSP-20	Data center and large enterprise networks
Cisco ASA 5585-X SSP-40	Data center and large enterprise networks
Cisco ASA 5585-X SSP-60	Data center and large enterprise networks
Cisco ASA Services Module	Data center and large enterprise networks
Cisco ASA 1000V Cloud Firewall	Data center and large enterprise networks

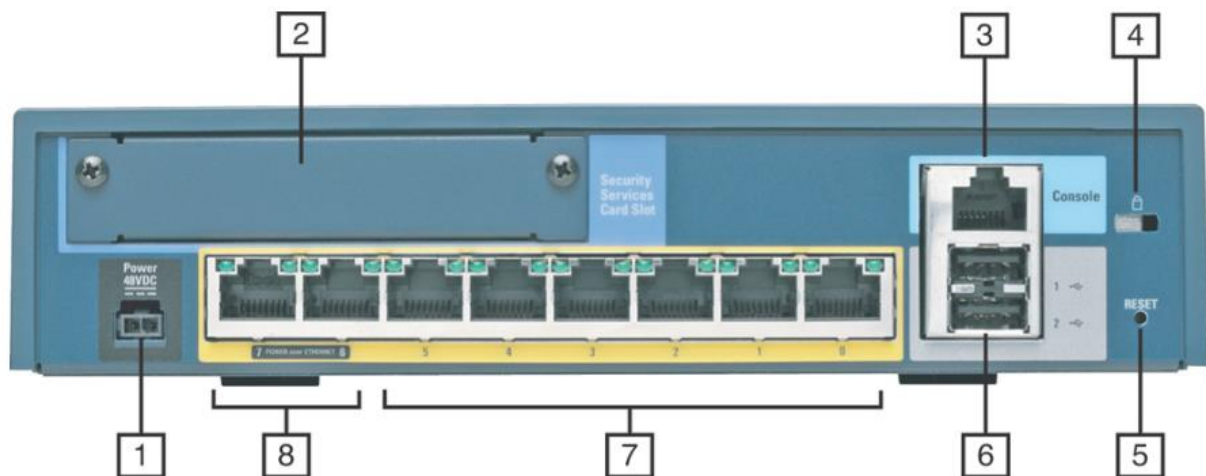
Software Versions:

7.0	7.1	7.2	8.0	8.1	8.2	8.3	8.4	8.5	8.6
8.7	9.0	9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8

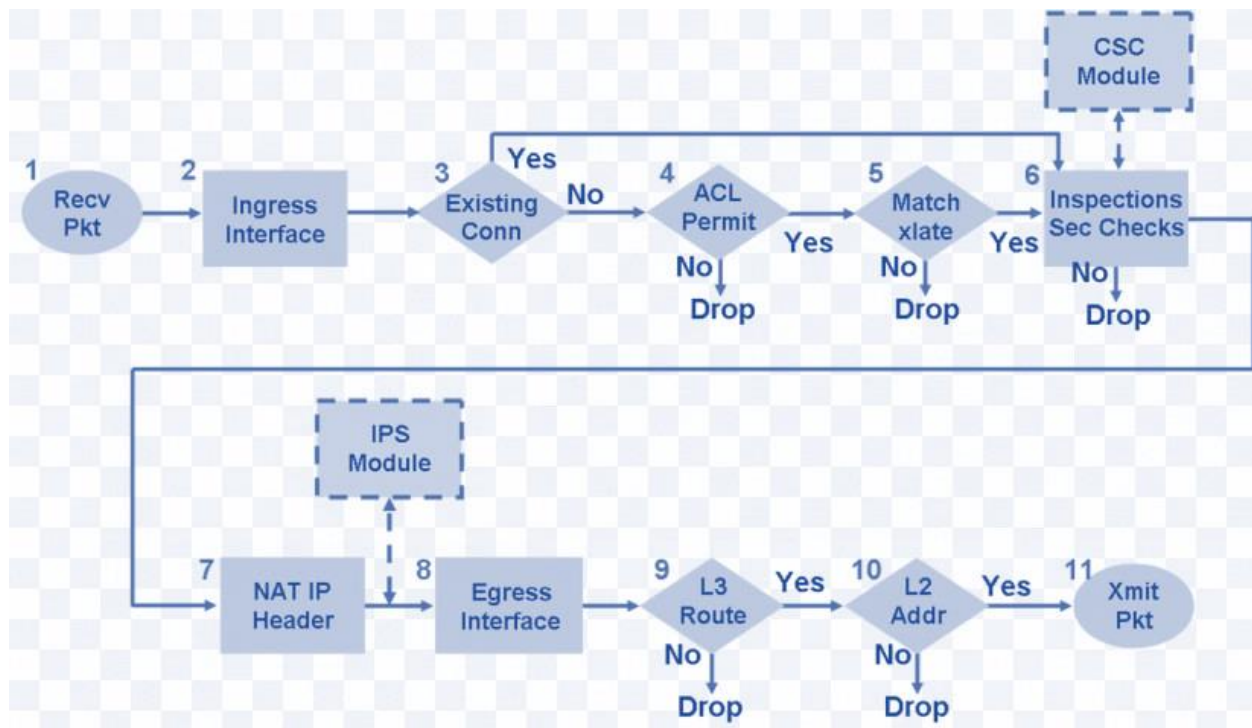
ASA Type/Model	ASA Software Version
Cisco ASA 5500 Series (5505, 5510, 5520, 5540, 5550, 5580)	ASA Version 8.4(x)
Cisco Catalyst 6500 Series ASA Services Module	ASA Version 8.5(x)
Cisco ASA 5500-X Series (ASA 5512-X, ASA 5515-X, ASA 5525-X, ASA 5545-X, and ASA 5555-X)	ASA Version 8.6(x)
ASA 1000V cloud and virtual firewall	ASA Version 8.7(x)
All product series ASA 5500, ASA 5500-X and 6500 Service Module	ASA Version 9.0(x)



1	USB 2.0	4	Flashing Green: Booting Solid Green: Booted Amber: Error
2	100 Mbps	5	Green: Unit is forwarding traffic Amber: Unit is on standby
2	Link Activity	6	Solid Green: VPN tunnel is established Flashing Green: VPN tunnel initiated Amber: VPN tunnel failed to initiate
3	Unit is receiving power	7	Security Services Card slot is occupied



1	Power connector	5	Reset button
2	Security Services Card (SSC) slot	6	USB 2.0
3	Serial console cable port for managing the device	7	Ports e0 through e5 are also Fast Ethernet ports (100 Mbps), but are not PoE enabled
4	Security lock slot	8	Ethernet ports e6 and e7 are Fast Ethernet ports (100 Mbps) and are PoE (Power over Ethernet) enabled



Stateful Packet Inspection



Stateful packet inspection looks at the **header** and **footer** of a packet.

Deep Packet Inspection



Deep packet inspection examines the **data part** of a packet.